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Quantitative and unbiased analysis of directional persistence in cell migration

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The mechanism by which cells control directional persistence during migration is a major question. However, the common index measuring directional persistence, namely the ratio of displacement to trajectory length, is biased, particularly by cell speed. An unbiased method is to calculate direction autocorrelation as a function of time. This function depends only on the angles of the vectors tangent to the trajectory. This method has not been widely used, because it is more difficult to compute. Here we discuss biases of the classical index and introduce a custom-made open-source computer program, DiPer, which calculates direction autocorrelation. In addition, DiPer also plots and calculates other essential parameters to analyze cell migration in two dimensions: it displays cell trajectories individually and collectively, and it calculates average speed and mean square displacements (MSDs) to assess the area explored by cells over time. This user-friendly program is executable through Microsoft Excel, and it generates plots of publication-level quality. The protocol takes ~15 min to complete. We have recently used DiPer to analyze cell migration of three different mammalian cell types in 2D cultures: the mammary carcinoma cell line MDA-MB-231, the motile amoeba *Dictyostelium discoideum* and fish-scale keratocytes. DiPer can potentially be used not only for random migration in 2D but also for directed migration and for migration in 3D (direction autocorrelation only). Moreover, it can be used for any types of tracked particles: cellular organelles, bacteria and whole organisms.

INTRODUCTION

Migration allows cells to efficiently explore their territory or the organism they belong to. The efficiency of cell migration depends on two essential parameters: speed and directional persistence. At the cellular level, directional persistence depends on sustained lamellipodial protrusion and cellular tail stabilization^{1,2}. Whereas cellular speed is relatively easy to quantify, directional persistence is not.

Development of methods to model cell migration

Early studies that attempted to mathematically model cell migration in isotropic medium assumed that a cell moves like a Brownian particle^{3,4}. Namely, a cell was assumed to undertake a persistent random walk (PRW), in which it moves directionally at short time intervals, but it loses its persistence at longer time intervals³. The time to cross from the persistent regime to the random one is the directional persistence time, usually designated as P . To obtain an estimate of P and other migration parameters, early studies fitted the MSDs of cells' trajectories to the Fürth's formula (equation 1), where D is the diffusion coefficient and t is time.

$$\text{MSD}(t) = 4D \left(t - P \left(1 - e^{-\frac{t}{P}} \right) \right) \quad (1)$$

This fitting was originally done with ordinary nonlinear least-squares regression analysis³. Later on, a more accurate fitting approach developed, which was based on generalized regression⁵. It was also noted that deviations for large time interval data were so big that perhaps cells did not move according to the PRW mechanism⁵.

Indeed, later studies have questioned the underlying assumption that migrating cells undergo PRW. For example, a study

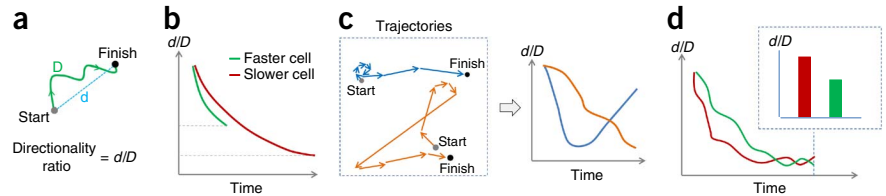
on migration of MDCK-F cells showed that the dynamics of these cells are anomalous in the sense that they deviate from the predictions of PRW⁶. The motile amoeba *D. discoideum* was reported to maintain directional persistence by yet another mechanism: zig-zagging⁷. Recent studies using mammary epithelial cell line MCF-10A modeled cell migration according to a two-phase mechanism similar to that of bacterial runs and tumbles: mammalian migration trajectories were segmented into two alternating modes, the directional mode and the reorientation mode⁸. This categorization was based on a user-defined threshold in turn angle, which is applied in the context of a scoring algorithm over three consecutive frames of a trajectory. Thus, the time spent in each mode could serve as a metric for directional persistence. Overall, a variety of sophisticated methods have been developed to describe cell migration in general and directional persistence in particular. However, most of these methods are computationally intensive and somewhat difficult for cell biologists to adopt.

In many instances, cell migration is biased by external cues, such as chemoattractants, extracellular matrices or electrical fields⁹. These external cues are thought to impinge on an intrinsic cell program of directional persistence: randomly moving cells are endogenously more or less persistent depending on the strength of their intrinsic positive-feedback loops^{1,10}. External cues would result in increased directional persistence when cells move toward the correct destination. Directional persistence is therefore fundamental to cell migration and crucial to quantify, regardless of whether the migration is guided by external cues or not.

We recently reported a novel protein, Arpin, which antagonizes an intrinsic positive feedback loop sustaining lamellipodial protrusion¹. As a consequence, Arpin promotes turning. When studying the effect of Arpin in cell migration, we assessed the different indices proposed to reflect directional persistence, namely directionality

Figure 1 | Directionality ratio and its pitfalls.

(a) Definition of directionality ratio. (b) For most cell types, directionality ratio decays quickly when measured over time. When directionality ratio is only measured in the last point of trajectory, speed can become a confounding factor if cells are not tracked for the same length of time. The faster cell, represented by a green curve, is likely to be tracked for a shorter time, as faster cells usually leave the microscope's field of view earlier than slow-moving cells. This will exaggerate the directionality ratio. (c) Even when it is reported over time, directionality ratio can be influenced by cell speed. In this example, the direction vectors for both blue and orange trajectories follow exactly the same angles; however, their step lengths, which reflect cell speed, are different. This results in quite different d/D curves, which are subsequently averaged to produce a noisy average curve, as depicted in d. Direction autocorrelation analysis would produce exactly the same result for these two trajectories. A real data example of noisy curves is shown in **Figure 2a**. (d) Even when cells are tracked over the same length of time, reporting directionality ratio in the last point can be artifactual owing to noise, especially in the high end of the time range, where d/D is low. In this example, the cells of the 'green' condition are more directional than those of the 'red' condition for a majority of time, except toward the end where noise sets in. However, if measured in the last point, green cells appear as less directional (inset).



ratio, MSD and direction autocorrelation. We found that all three indices are useful under different circumstances.

The directionality ratio, also sometimes referred to as the straightness ratio, is the preferred method of cell biologists. This parameter is the straight-line length between the start point and the endpoint of the migration trajectory, divided by the length of the trajectory (**Fig. 1**). This ratio is equal to 1 for a straight cell trajectory and approaches 0 for a highly curved trajectory. Thus, the directionality ratio is easy to understand and compute.

The MSD is often used in physics to relate displacement of objects. MSD portrays average square displacements over increasing time intervals between positions of a migration trajectory (**Fig. 2**). In the context of cell migration, MSD is a good measure of the surface area explored by cells over time, which relates to the overall efficiency of migration. MSD carries information about both speed and directional persistence, and it is often expressed as a log-log plot, with $\log(\text{MSD})$ on the y axis and $\log(\text{time interval})$

on the x axis. The slope of these log-log plots, often called the α -value, is a handy index for directional persistence: it equals 1 for randomly moving cells and 2 for cells that move in a perfectly straight manner.

The direction autocorrelation belongs to the wide family of spatial autocorrelation indices widely used in diverse disciplines, such as physics, geography and economics¹¹. Autocorrelation measures how a certain quantity correlates with itself over different scales. In our case, this quantity is the angle of migration that displacement vectors form, measured over different timescales (**Fig. 3**). The advantage of direction autocorrelation is that it is not influenced by speed, but rather it measures only how angles describing the trajectory are aligned with each other. Despite this, its use in cell migration¹² has been limited, mainly owing to its relative computational difficulties.

In this protocol, we describe a suite of computer programs called DiPer (**Table 1**), which we developed during our study of Arpin¹ to calculate these different indices reflecting directional persistence (**Figs. 2–4**). In addition, DiPer provides auxiliary programs to plot trajectories individually and collectively, to calculate cellular speed and to reduce noise on centroid positions (**Fig. 5**). Our programs are easy-to-use Excel macros: most researchers are already familiar with Microsoft Excel and many use this software to organize their migration data. Microsoft Excel works in conjunction with a built-in application, the Visual Basic Editor: the program code is

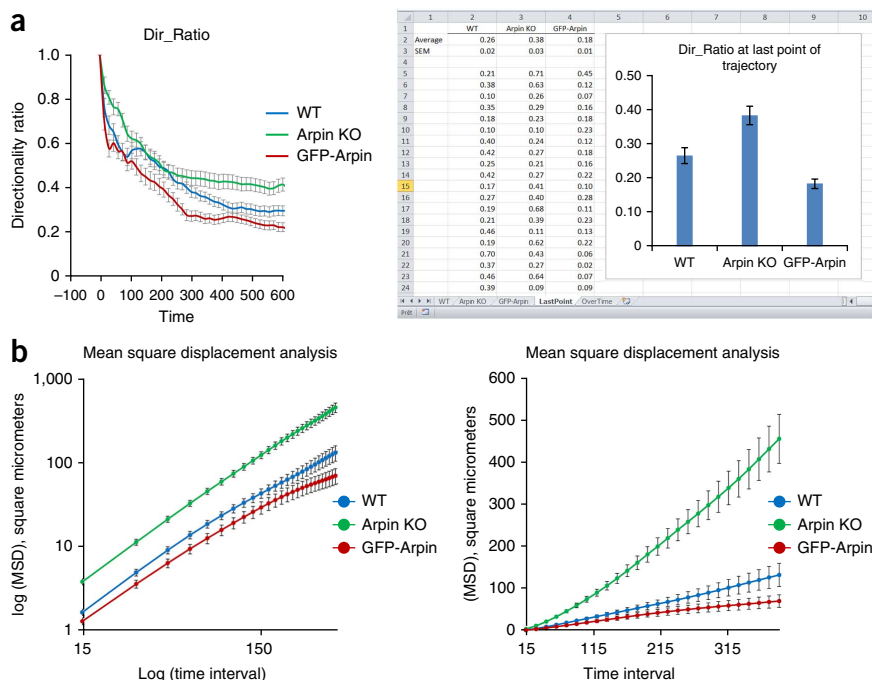
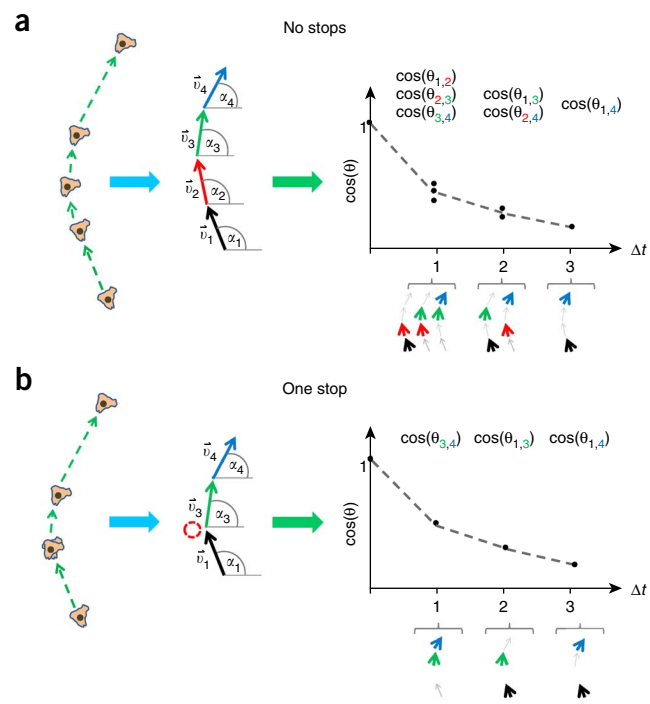


Figure 2 | Outputs of DiPer programs obtained from *D. discoideum* trajectories (**Supplementary Data 4**) for wild-type (WT), Arpin knockout (KO) and GFP-Arpin rescue amoeba. (a) Dir_Ratio, showing directionality ratio over elapsed time, left, and in the last point of trajectory, right. (b) MSD, plotted on a log-log scale, left, and on a linear scale, right. Minor manual adjustments of graphs were performed in Excel before exporting the image files. $N = 42$, 38 and 42 trajectories for WT, KO and GFP-Arpin, respectively. Error bars are s.e.m.

Figure 3 | Analysis of direction autocorrelation. Displacement vectors describing migration trajectory are normalized to the same length, and vector angles (α) are compared pairwise over different time intervals Δt by computing the angle difference (θ). The correlation coefficient is the cosine of the angle difference. For each time interval, the vectors used to compute the coefficients are shown below the axis, and the cosines are listed above the plot. In this hypothetical trajectory comprising 4 displacements, the maximal number of possible time intervals is 3, and the number of vector comparisons decreases from 3 to 1 as time increases. **(a)** In this situation, the cell migrates without stops, and both autocorrelation programs that we provide, Autocorrel and Autocorrel_NoGaps, will produce identical results. **(b)** When the cell stops, in this case from position 2 to position 3, vector $\vec{v}_2$ cannot be produced (red circle). The use of Autocorrel program will skip $\vec{v}_2$ for each time interval, as shown below the x axis. Autocorrel_NoGaps will produce a very small random vector to avoid generating a gap, and hence it will return to the situation described in **a**.



pasted into Visual Basic Editor, from which the program is run, producing the desired result (Fig. 6). In addition to displaying the resulting plots, DiPer also displays data corresponding to all the intermediate calculation steps, which makes our programs easy to understand and to correlate to the provided open-source code. As our computer programs are annotated, users can modify our code to suit their particular needs. Moreover, these programs output high-quality plots, which may be used for presentations or publications. An abridged version of the workflow described in the PROCEDURE is schematically presented in Figure 6.

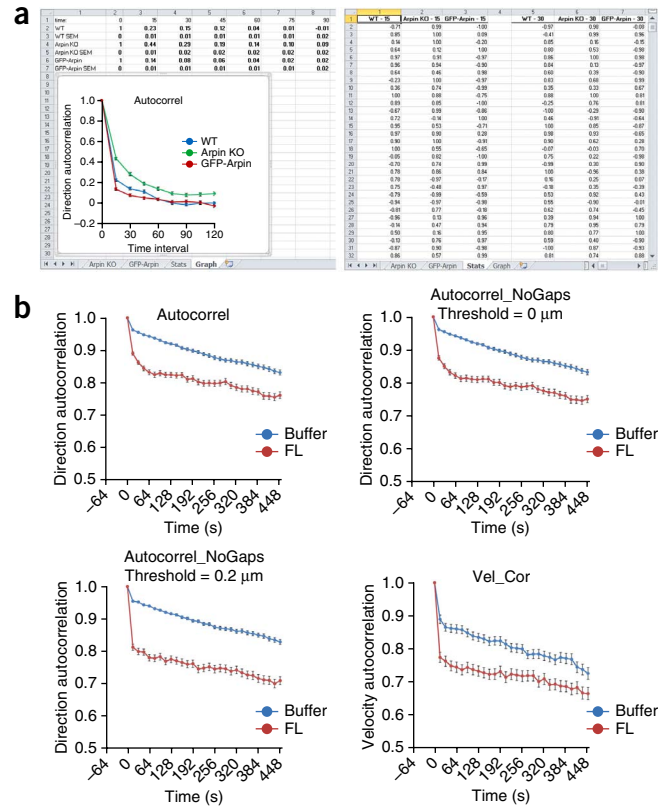
Several cell migration software packages exist, and they specialize in different aspects. For example, the Cell_Motility software provides the user with MSD analysis with either overlapping or nonoverlapping time intervals, as well as persistence time and r.m.s. speed derived from the Fürth's formula¹³. Other programs,

such AVeMap and Pathfinder, focus on migrations of groups or clusters of cells^{14,15}. However, to our knowledge, no other software exists that computes direction autocorrelation, as DiPer does through an environment that is familiar to the user: Microsoft Excel. It is more advantageous to use DiPer when one wants to measure directional persistence over all possible timescales and uncoupled from the effects of speed.

TABLE 1 | Computer programs comprising DiPer.

File/program name	Function	Supplementary Data file number
Plot_At_Origin.txt	For each condition (worksheet), plots all cell trajectories emanating from the origin	1
Make_Charts.txt	Plots each trajectory in a separate chart next to the corresponding xy coordinates	2
Sparse_Data.txt	Deletes frames from migration trajectories, leaving 1 out of each N frames. User specifies N	3
Speed.txt	Computes and plots average speed for each condition and reports cell number	6
DirRatio.txt	Plots the directionality ratio over time as well as in the last point. Displays these in separate new sheets	7
MSD.txt	Performs the MSD analysis and plots on a log-log scale. Provides MSD coefficients for user to compute α -values	8
Autocorrel.txt	Performs the direction autocorrelation analysis and plots the resulting curves. Provides stacked ranges of coefficients for each time interval	9
Autocorrel_NoGaps.txt	Same as Autocorrel above, except, instead of omitting coefficients for repeat coordinates, it generates a new vector, thus precluding the gaps	10
Autocorrel_3D.txt	Same as Autocorrel above, except performs direction autocorrelation analysis in three dimensions (xyz)	11
Vel_Cor.txt	Performs the normalized velocity autocorrelation analysis	12

Figure 4 | Outputs of DiPer autocorrelation programs for automatically tracked *D. discoideum* cells and manually tracked fish keratocytes (FK) cells. **(a)** Automatically tracked *D. discoideum* cells. Left, Autocorrel produces a plot with direction autocorrelation curves displayed in worksheet 'Graph'. Right, autocorrelation coefficients used for computing averages are listed in worksheet 'Stats'. Here the first cluster of columns corresponds to the first time interval at 15 min, and the second cluster corresponds to the second time interval at 30 min. $N = 42, 38$ and 42 trajectories for WT, KO and GFP-Arpin, respectively. Error bars are s.e.m. **(b)** For manually tracked FK cells microinjected with either buffer (Buffer) or full-length Arpin (FL), DiPer outputs are shown for programs Autocorrel (top left), Autocorrel_NoGaps (top right, zero threshold), Autocorrel_NoGaps (bottom left, $0.2 \mu\text{m}$ threshold) and Vel_Cor (bottom right). Trajectories for cells injected with Arpin FL contain substantially more stops and turns than buffer-injected cells. Hence, FL curves decay faster than buffer curves. The use of a zero distance threshold for Autocorrel_NoGaps approximates the output of Autocorrel more faithfully than using a nonzero ($0.2 \mu\text{m}$) threshold, because the latter artificially depresses the FL curve owing to a higher preponderance of randomly oriented vectors in place of gaps. Vel_Cor program output reports normalized velocity autocorrelation function, which approximates direction autocorrelation. For all four graphs, there was a statistical difference between means of Buffer and FL for each time interval ($P < 0.01$, Mann-Whitney ANOVA). Minor manual adjustments of graphs were performed in Excel before exporting the image files. $N = 16$ and 15 trajectories for buffer-injected and FL-injected keratocytes, respectively. Error bars are s.e.m.



Calculations used by DiPer

$$\text{Directionality ratio} = \langle d_t / D_t \rangle_C \quad (2)$$

The directionality ratio is computed as an average for a population of C cells according to equation 2, and it is plotted against elapsed time t (Figs. 1 and 2a). d_t denotes the straight-line distance between the start point of the trajectory and the current position at time t , whereas D_t is the actual length of trajectory between the start point and the current position. Brackets indicate averaging for each time point over C cells.

$$\text{MSD}(n) = \frac{1}{N-n+1} \sum_{i=0}^{N-n} [(x_{(i+n)\Delta t} - x_{i\Delta t})^2 + (y_{(i+n)\Delta t} - y_{i\Delta t})^2] \quad (3)$$

MSDs are computed according to equation 3, by using overlapping time intervals (Fig. 2b). Please see the 'Limitations' section for a discussion of overlapping versus nonoverlapping time intervals. $\text{MSD}(n)$ is the MSD for a given cell for step size n , where N is the total number of displacements per trajectory. Δt is the minimal time interval between adjacent points in the trajectory, which is defined by the frame acquisition rate. To obtain a population average for each time interval $n\Delta t$, MSDs are averaged over all cells, and this average is plotted against time interval. The average is not weighted for trajectory time length, because the main purpose is to provide α -values, which are calculated from the low end of the time interval range. Please see the 'Limitations' section for a discussion of α -values.

$$\text{DA} = \frac{1}{N-n+1} \sum_{i=0}^{N-n} (\vec{v}_{(i+n)\Delta t} \cdot \vec{v}_{i\Delta t}) = \frac{1}{N-n+1} \sum_{i=0}^{N-n} (\cos(\alpha_{(i+n)\Delta t} - \alpha_{i\Delta t})) \quad (4)$$

$$\langle \text{DA} \rangle_C = \sum_{j=1}^{j=C} (\text{DA})_j \cdot N_j / \sum_{k=1}^{k=C} N_k \quad (5)$$

Direction autocorrelation coefficients are computed with overlapping time intervals according to equation 4, where DA is the

average direction autocorrelation coefficient for a given cell at step size n , and N is the total number of displacements (Fig. 4). We compute DA as the scalar product of normed vectors $\vec{v}_{i\Delta t}$ and $\vec{v}_{(i+n)\Delta t}$. Δt is the minimal time interval between adjacent points in the trajectory, which is defined by the frame acquisition rate. The angle at each time point of trajectory is α . To obtain a population average per given time interval for C cells, we averaged DA coefficients according to equation 5, so as to account for different time length of trajectories; thus, longer trajectories contribute proportionately more autocorrelation coefficients than shorter ones.

$$\text{Norm} = \frac{1}{N} \sum_{i=0}^{N-1} |\vec{v}_i|^2 =$$

$$= \frac{1}{N * (\Delta t)^2} \sum_{i=0}^{N-1} [(x_i - x_{i+1})^2 + (y_i - y_{i+1})^2] \quad (6)$$

$$v_{ac}(n) = \frac{1}{N-n} \left(\sum_{i=0}^{N-n} \vec{v}_i \cdot \vec{v}_{i+n} \right) * \frac{1}{\text{norm}}$$

$$= \frac{1}{N-n} \sum_{i=0}^{N-n} \left[\frac{(x_i - x_{i+1})(x_{i+n} - x_{i+n+1}) + (y_i - y_{i+1})(y_{i+n} - y_{i+n+1})}{(\Delta t)^2} * \frac{1}{\text{Norm}} \right] \quad (7)$$

Normalized velocity autocorrelation function is computed for N positions of a trajectory, with Δt as the minimal time interval between adjacent points in the trajectory, and step size n . First, a normalization factor (Norm) is computed for adjacent displacements, where $\vec{v}_i$ is the velocity vector at position i with start coordinate (x_i, y_i) and end coordinate (x_{i+1}, y_{i+1}) (equation 6). Then, Norm is used to compute velocity autocorrelation coefficient $v_{ac}(n)$ for each step size n , according to equation 7.

Figure 5 | Outputs of DiPer auxiliary programs obtained from *D. discoideum* trajectories for wild-type (WT), Arpin knockout (KO) and GFP-Arpin rescue amoeba. See **Supplementary Data 4** for the data analyzed. (a) Plot_At-Origin. Brightness was enhanced in Adobe Photoshop. $N = 42$, 38 and 42 trajectories for WT, KO and GFP-Arpin, respectively. Error bars are s.e.m. (b) Make_Charts output of the first three trajectories for WT. (c) Speed. (d) Sparse_Data, with original data on the left and sparsed data to the right of the arrow, in which two out of three frames are removed. The outputs for a–c were obtained from sparsed data. The data plotted in b,d are in the associated Source Data file.

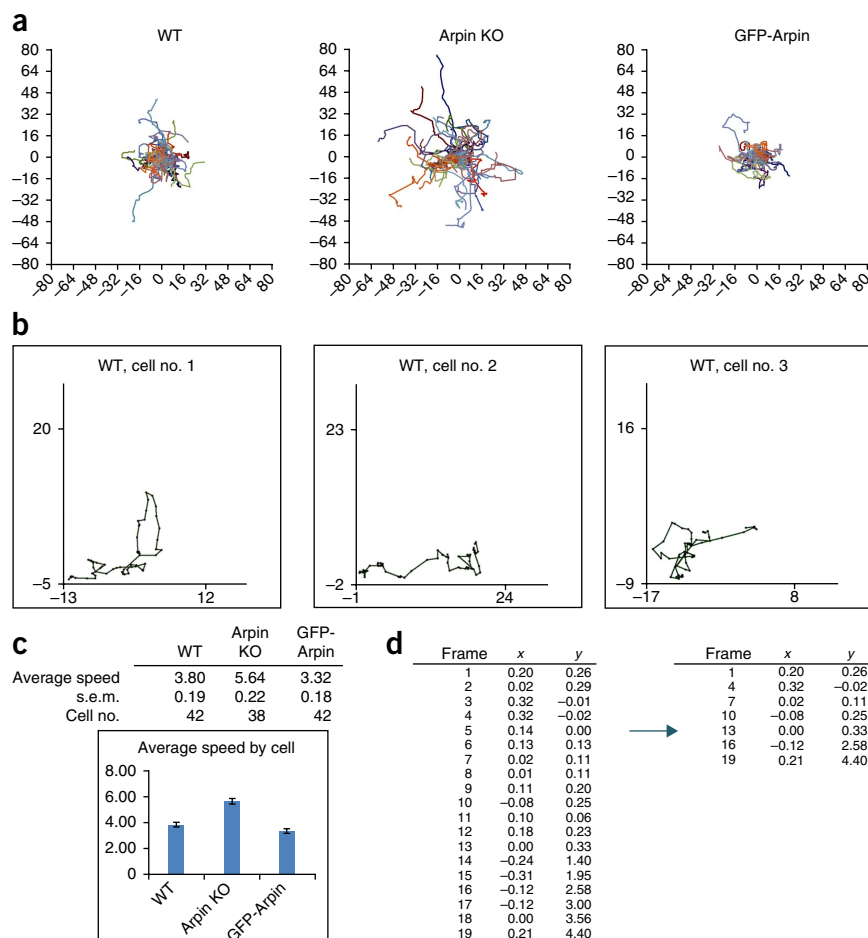
Averaging is performed as in equation 5, and the results are plotted against time interval (Fig. 4).

Experimental design

Before using DiPer programs to assess directional persistence, we recommend addressing some preliminary questions about the trajectory data. How were the cells tracked: manually or automatically? Was the frame acquisition rate properly chosen? How noisy are the data? In this section, we briefly discuss the implications of such questions and show how auxiliary DiPer programs can help ascertain trajectory quality before assessing directional persistence.

Manual versus automatic tracking of cells can have an important effect on consequent data analysis. When cells are tracked manually, the user clicks on the geometric center of the cell or on the nucleus, and the xy coordinate corresponding to the clicked pixel is recorded. This tracking method has the advantage of being more reliable than automatic tracking, which can record artifacts. However, manual tracking is labor-intensive and it is prone to the ‘grid-effect’: certain angles between displacements (0, 45, 90, 135, 180 and so on) will be overrepresented. In addition, if a cell idles from one frame to the next, the user will probably click on the same pixel, producing a repeat coordinate. This has implications for the direction autocorrelation analysis, as discussed below. For manual tracking, we use an ImageJ plugin known as MTrackJ¹⁶, which produces tracking data in a format that can be readily run on DiPer.

In contrast to manual tracking, in automatic tracking the geometric centroid is computed by the tracking software and is usually a unique number, which precludes repeat coordinates. The disadvantage, though, is that it may also record tiny (sub-pixel) displacements between frames that are unrelated to the motion along the trajectory, but rather reflect cell shape changes or movement of the nucleus within the cell. This is known as the ‘positioning error’⁵, and it is more likely to occur if the frame rate is set too high. In addition, automatic cell tracking is more prone to record artifacts such as floating debris, dead cells or dividing cells. Therefore, we recommend to first visually examine



the entire cohort of cell trajectories using our Plot_At-Origin program (Supplementary Data 1), which displays all trajectories emanating from the origin, thus providing a bird’s-eye view (Fig. 5a). These floating artifacts will appear as unreasonably large jumps, which can be spotted on the plot and corrected. When one finds abnormalities such as large jumps, one may choose to more closely examine individual trajectories, which may be obscured in a collective plot by other trajectories of the same cohort. For this purpose, we provide a program called Make_Charts (Supplementary Data 2), which displays individual trajectories next to corresponding xy coordinates (Fig. 5b). The output graphs of both programs Make_Charts and Plot_At-Origin can be readily used in publications, as articles on cell migration usually display trajectories individually and collectively. When using Plot_At-Origin to compare between two or more conditions, an approximately equal number of cells filmed over the same length of time must be shown for each condition to assure a valid visual comparison.

Choosing a correct frame acquisition rate for your microscopy experiment is an important step before tracking and data analysis. Frames should be acquired often enough not to miss small cell displacements, but not so often as to record noise. However, it is a safer strategy to acquire more frequently rather than less, because it is much easier to then skip extra frames than to repeat an entire experiment. For this reason, we provide here a program called Sparse_Data (Supplementary Data 3), with which the user can

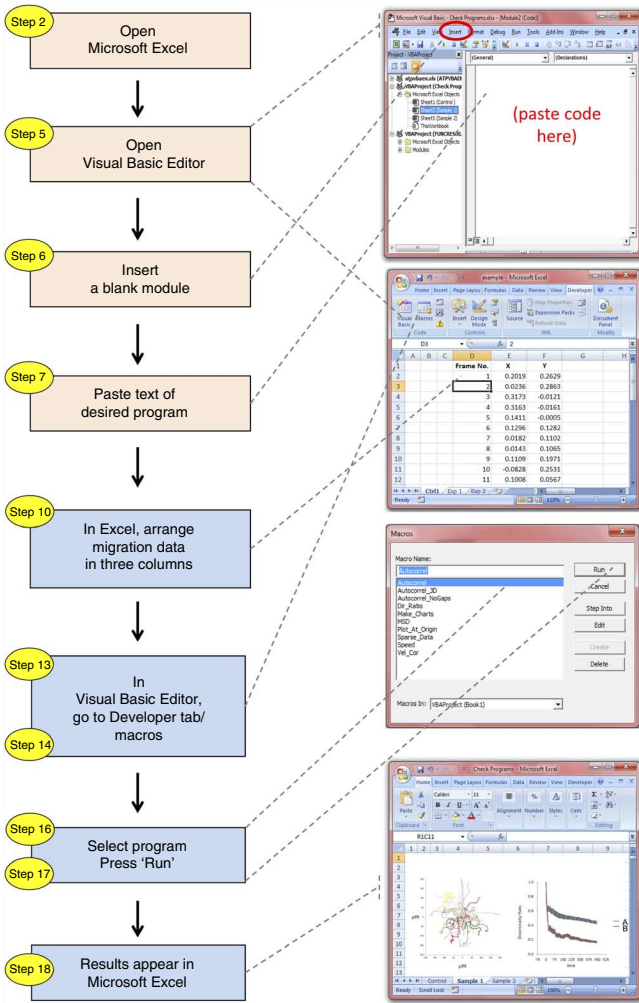


Figure 6 | Flowchart of DiPer procedure, highlighting the key steps in the column on the left. Steps 2–7 in light beige rectangles encompass the installation of DiPer on your computer. Steps 10–18 encompass running DiPer programs and obtaining results. For visual representation, the right column contains screen shots for key steps in the procedure, with dashed lines pointing to details.

specify the number of frames to skip per moving window (Fig. 5d). For example, the graphic outputs of DiPer shown in Figures 2 and 5 are from migration data of amoeba *D. discoideum* acquired every 5 s that was processed with Sparse_Data; for a window size of 3, every two out of three time frames were skipped, thus leaving frames 1, 4, 7, 10 and so on, so that the new minimal time interval is 15 s. We find that a good way to gauge the correct interval is to observe actual movies of cell migration and to correlate them with recorded trajectories.

Limitations of DiPer

Directionality ratio. The two major advantages of directionality ratio are easy calculation and easy interpretation. However, these advantages are counterbalanced by biases, which can warrant the use of methods other than this simple index. Directionality ratio is usually calculated between the start and end of each cell trajectory (Fig. 1a). However, this index varies markedly over the time course of a cell trajectory (Fig. 2a). Any cell migrating at random

is directional at short timescales, typically the time during which it remains polarized in a single direction, but it loses directional persistence at longer timescale, as no external cues bias migration direction in random migration assays. As a consequence, the directionality index decays over time. Unfortunately, fast-moving cells also tend to escape the microscopic field of view earlier than control cells, and thus they tend to generate shorter trajectories. This effect induces the directionality ratio to be calculated for the different conditions at different time points, hence strongly biasing this index of directional persistence by speed (Fig. 1b). The representation of directionality ratio over time, which our program provides, is a better proxy for directional persistence than the end-point index. Because of the influence of speed, the directionality ratio curves for individual trajectories can vary significantly (Fig. 1c), which can result in a noisy average curve, especially when small cell populations are assayed (Fig. 1d). In addition, if cells are tracked for different lengths of time, this noise may become amplified in the latter time points of the average curve, because fewer cells with long trajectories contribute to these latter time points (Fig. 1d).

MSD. This section on MSD and the next one on direction auto-correlation deal with methods that depict directional persistence over increasing time intervals. In particular, in the case of limiting trajectory data, the resulting plots for both techniques may be misleading and must be interpreted with caution. As the time interval increases, there is also an accompanying decrease in the number of MSD coefficients available for averaging. This increases the uncertainty of the reported average for the high time interval data. DiPer calculates averages using overlapping time intervals between frames, which are not statistically independent, in contrast to nonoverlapping intervals⁵. However, we chose the overlapping intervals method because the number of coefficients available for averaging is greater than in the nonoverlapping case, which can be an issue when data are limiting⁵. In addition, at shorter time intervals, averaging overlapping intervals is slightly more precise than with nonoverlapping method⁵, which is well-suited for calculation of α -values, as these are usually obtained from shorter time intervals.

As discussed earlier, cell migration rarely fits the PRW mechanism^{6–8}. Therefore, DiPer does not calculate persistence time and other parameters that stem from F rth's formula. Readers who wish to calculate these parameters are referred to Martens *et al.*¹³. Nonetheless, in addition to plotting MSD curves, DiPer provides the user with a list of MSD coefficients from which α -values can be easily computed using Excel's slope function. Please see the ANTICIPATED RESULTS section for more details. The α -value is the slope of MSD curves, which have been 'linearized' using the logarithm of the MSD and the logarithm of the time interval. Thus, the α -value can serve as a proxy for directional persistence, albeit with caution. When the logarithms of the MSD and time interval are taken, the resulting plots are not exactly linear, and they become noisier toward the longer time intervals. This necessitates user bias in computation of the slope: the user specifies an arbitrary number of points from the low end of the time interval range over which the slope is computed. One common method is to use a fixed percentage (e.g., 10%) of the low-end time intervals over which to compute the slope. This can be a percentage of a fixed number of points, for example, 10% of the time length of

the longest trajectory or of the average trajectory. Another way to compute the α -value is to iteratively increase the time interval while checking at each step that the Pearson correlation coefficient is above a certain value. This method presents yet another bias: different numbers of points are used for different cells. Therefore, we recommend using the fixed percentage method, which is also easier to compute than the second method.

Direction autocorrelation. The first step in direction autocorrelation analysis is to leave out all information about cell speed (i.e., to normalize the displacement vectors). This precludes any influence of speed on downstream calculations. The only parameter that matters is the angle that they form with respect to each other. For each pair of displacement vectors, autocorrelation coefficients are obtained as cosine of their angle difference, so as to yield a value of 1 when displacement vectors are parallel, a value of 0 when they are orthogonal and a value of -1 when they are antiparallel. Then, for each time interval these coefficients are averaged, and the averages are plotted against time intervals, thereby producing decaying curves (Fig. 3a). The rate at which these curves decay reflects a cell's propensity for turning (Fig. 4).

The accuracy of the direction autocorrelation method is limited by the reliability of each vector. More specifically, the direction of each vector is defined by its start and end positions, which depend on the frame rate. On the one hand, if the user sets the frame rate too low, true vectors will not be recorded. On the other hand, if the user sets the frame rate too high, spurious vectors will be recorded, resulting in autocorrelation curves that decay too quickly. In summary, the true direction of each velocity vector is not known, but rather approximated by the measured data points.

Another possible limitation of the direction autocorrelation method can arise when there is absolutely no displacement from one frame to the next, in which case a velocity vector cannot be generated (Fig. 3b). This occurs during manual tracking of cells owing to the pixel grid effect, as discussed above. So how to compute direction autocorrelation in case of zero displacement? We propose several options with DiPer. The first one, our default direction autocorrelation program called Autocorrel, simply omits the 'missing' velocity vector, thus creating a data gap. This gap results in two missing autocorrelation coefficients at each time interval of the analysis (Fig. 3b). These gaps are not problematic if sufficient data (long trajectories or high number of cells) are available. Indeed, Autocorrel is our program of choice, because it does not create any artifacts that arise when including or specially treating the repeat coordinates.

In the case of insufficient measurements, we provide a second version of direction autocorrelation called Autocorrel_NoGaps. This version prevents gaps by adding a very small random number to the repeat coordinate. This addition creates a tiny velocity vector, less than a pixel long, which prevents missing values from occurring at all time intervals. The user may also choose to set a distance threshold on idling events, that is, to define a distance below which the cell is considered to be idling. In Figure 4b, we provide two outputs of Autocorrel_NoGaps, one with a zero

threshold and another with a higher threshold. We note that Autocorrel_NoGaps can create artifacts such as downward-facing kinks at certain time intervals or depression of entire curves when many idling events are present. These artifacts can be magnified with increasing threshold. Please see Figure 4b for more details. Therefore, the use of Autocorrel_NoGaps should be avoided when Autocorrel may be used instead.

Another way to avoid inducing gaps in cases of zero displacement, and do so without specially treating repeat coordinates, is to use the normalized velocity autocorrelation analysis⁶, which is available in DiPer as program Vel_Cor. Velocity is a vector with the magnitude of speed and the direction of motion. Thus, by normalizing velocity by speed (equations 6 and 7), this technique approximates the direction autocorrelation analysis (Fig. 4b). In the literature, velocity autocorrelation has been used in migration of *D. discoideum* to report very large turns, corresponding to switches in direction¹⁷. Vel_Cor is robust against frame-to-frame stops, as are programs Dir_Ratio and MSD.

Migration in 3D environment has become routine for cell biologists studying cell migration in tissues and live organisms. Therefore, we provide here a computer program, Autocorrel_3D, which is designed for this. As 3D tracking is done automatically, it produces unique nonrepeating xyz coordinates, thereby precluding the problem of frame-to-frame stops. Hence, our Autocorrel_3D does not use a random-number approximation of the repeat coordinate.

Potential applications

Having been presented with three different methods for assessing directional persistence, the user may wonder when each method is applicable. In random migration assays, one assesses the propensity of a cell to move in any direction, in contrast to one particular direction. Thus, direction autocorrelation analysis is appropriate here, because this analysis indicates how well migration vectors are aligned with each other, and not necessarily toward a particular direction. Another major benefit of direction autocorrelation analysis is that it separates the influence of speed from directional persistence.

In contrast, in directed migration assays, for example toward a chemoattractant or a wound, it is important for the cell not only to crawl in the correct direction but also to take bigger steps when moving in the correct direction. Therefore, direction autocorrelation is less appropriate here, because it negates any influence of speed. However, both the directionality ratio and MSD methods are appropriate, as these report both speed and directionality components.

Cell migration in the 3D environment of the organism can be also assessed with DiPer for directional persistence using the Autocorrel_3D program. For example, the collective migration of prechordal plate cells to the animal plate in the zebrafish embryo is a useful tool in studying *in vivo* effects of cell migration in 3D (ref. 1). The use of DiPer programs is not limited to individual cells, and it may also be applicable to whole organisms. A variety of organisms crawl and turn in both individual and social contexts, from ciliated protozoa¹⁸ to the larva of the common fruit fly *Drosophila melanogaster*¹⁹.

MATERIALS

EQUIPMENT

Data

- Cell trajectories as position coordinates with corresponding frame numbers, saved as a Microsoft Excel file. Step 10 gives guidance on exact formatting of the data. **Supplementary Data 4** and **5** are examples of input trajectories for *D. discoideum* and fish keratocytes, respectively

Computer

- Any PC running Microsoft Excel version 2007 or later

Software

- Program source codes are available as text files as supplementary information (**Supplementary Data 1–12**)

PROCEDURE

Install DiPer on your computer

- 1| Download the program codes from **Supplementary Data 1–3** and **6–12**. Save these text files on your computer.
- 2| Open Microsoft Excel.
- 3| If your Ribbon has no Developer tab, activate it. To do this, choose 'Popular' from Excel Options, check the box next to 'Show Developer tab in the Ribbon' and click 'OK'.
- 4| In the Developer tab, click on 'Macro Security'. A window called 'Trust Center' will open.

Select 'Enable all Macros', check the box next to 'Trust access to the VBA project object model' and click 'OK'.

? TROUBLESHOOTING

- 5| In the Developer tab, click on 'Visual Basic' (shortcut Alt+F11). A screen called 'Microsoft Visual Basic for Applications' will appear. This is the Visual Basic Editor.
- 6| At the top of the Visual Basic Editor, click on 'Insert' and select 'Module'. A blank pane (module) will appear. This is the area into which the programs will be pasted.
- 7| Copy a program (text file) from Step 1 and paste it into the blank module.
▲ CRITICAL STEP To avoid copying errors, do not use the cursor for selecting. Instead, select all (Ctrl+A), then copy (Ctrl+C) and then paste (Ctrl+V).
- 8| Repeat Steps 6 and 7 for the remaining nine programs, or a desired subset of them. Further information about the content of each program can be found in **Table 1**.
- 9| Save as a Macro-Enabled Workbook, naming your file (e.g., 'Empty File.xlsm'). This macro-enabled format saves the computer programs, i.e., the macros you just pasted, to the Excel file. At this point, your workbook does not contain any trajectories, and it may be used in the future to insert new trajectories. We refer to this as the 'Empty File'.
▲ CRITICAL STEP If you e-mail a macro-enabled workbook, antivirus software may delete the associated macros.

Arrange your data

- 10| Arrange your migration trajectory data according to the following format. Each worksheet in the workbook must correspond to one condition. For example, if you have one control condition and two experimental conditions, your workbook should contain three labeled worksheets (e.g., control 1, sample 2 and sample 3). To label worksheets, double-click on the worksheet tab (at the bottom of screen) and type. Each worksheet must contain all trajectories for that condition listed one immediately after the other without gaps. That is, do not insert empty rows between trajectories. If your workbook contains empty sheets, delete them by right-clicking and selecting 'Delete'.

Specific columns should contain the following information:

Column	Content
1–3 or A–C	These are not used by the program. They can either be left blank or filled with identifying information such as experiment number, sample number or date
4 or D	Must contain the frame number of the trajectory, which must increase monotonically down the page. When a given cell trajectory ends and a new one begins, the frame number will decrease to 1, allowing DiPer to detect the start of a new trajectory
5 or E	Must contain the <i>x</i> coordinate of your tracked cell or particle
6 or F	Must contain the <i>y</i> coordinate of your tracked cell or particle
7 or G	If your data is in three dimensions, paste <i>z</i> coordinate data in into column no. 7 (G)

▲ CRITICAL STEP In some cases, when trajectory data are pasted into the columns, Excel does not recognize the data as numbers. Please see the TROUBLESHOOTING section for ways to remedy this problem.

? TROUBLESHOOTING

11| (Optional) Save your file under a new name, such as ‘Exp 1.xlsm’. We suggest keeping this original file as is and not running any programs on it. We refer to this file as the ‘Original File’.

12| (Optional) Save the Original File for a second time under a new name. We refer to this second file as the ‘Copy File’. You will run programs on the Copy File (see below).

Run your program

13| Click on the Developer tab (**Fig. 6**).

14| Click on the ‘Macros’ button (**Fig. 6**). You will see a screen appear with a list of the programs that you have inserted.

15| Select a program from the list (**Fig. 6**). Select only one of the programs listed in **Table 1**.

16| Press ‘Run’. The status bar at the bottom left of the screen will display ‘Please wait ...calculations in progress’. When the program finishes, the status bar will display ‘Ready’. If an error occurs, please see the TROUBLESHOOTING section.

? TROUBLESHOOTING

17| Some programs will ask you to input parameters, such as time interval between frames. Please type only numbers and not the units. (The latter can be typed into the axes labels of the graphs.) Press ‘OK’ or ‘Enter’.

18| Your results should now be displayed in Microsoft Excel. All the programs except Plot_At_Origin display plots in a new worksheet represented by a new tab. Click on the tab to view your results.

19| (Optional) Save the resulting workbook under a new name. We suggest appending the name of your file with the name of the program, for example, ‘Exp1_autocorrel.xlsm’.

Rerunning the same program or running a different program

20| (Optional) To rerun the same program or a different program on same data, close the resulting file. Open the Copy File and repeat Steps 13–16 with a program of your choice.

! CAUTION Our programs make substantial changes to the workbook, such as displaying large amounts of data and inserting new worksheets. To avoid errors, we advise against running any programs on worksheets that are already displaying results, that is, on data that have already been run.

Formatting and exporting charts

21| (Optional) If you would like to change the appearance of your chart elements, such as line widths or axis colors, there are two ways: manually or programmatically. To change manually, right-click on the element you want to edit and choose ‘Format’. Excel will take you to formatting options. If you would like to change the format programmatically, please refer to the ANTICIPATED RESULTS section.

22| To save the resulting charts as high-quality graphs for publication or presentation, select the chart by clicking on it, click on the 'Office' button (top left of screen) and either save as a PDF or XPS file, or print as a PDF or XPS file.

? TROUBLESHOOTING

Step 4

When Excel does not recognize a piece of pasted data as a number, it will not be adjusted to the right margin of the rectangular cell in which the number appears. Instead, it will be adjusted to the left. The issue may be due to how your Microsoft Excel is configured to recognize decimal separators: a comma versus a period, or vice versa. To check how your system is configured, go to Excel 'Options' → 'Advanced' → 'Editing Options'. Next, go to the box 'Use system separators'. You will see what decimal separator is used, and you may change this as necessary. Alternatively, you may instead change your data by replacing all commas to periods or vice versa. This can be easily done using the Find and Replace function (shortcut Ctrl+H). In some extreme cases, you may need to replace a comma with a comma, or a period with the period, to get Excel to recognize your data as numbers.

Steps 10 and 16

If an error occurs while running code, Excel will tell you the error type and number. Common causes of errors are as follows: using a Macintosh computer rather than a PC; using Excel versions earlier than 2007; running a file than has already been run; incorrect formatting (Step 4); or only filling in one row for each trajectory. If the error is not due to these reasons, take note of it.

The easiest way to document an unusual error is by taking a screenshot using the 'PrtSc' button. Go to a basic image editing program, such as Paint, and press Ctrl+V, which pastes your screen image. Crop the error message box and save it as an image file; send the documentation of your error to the first author.

In Excel, the actual error box will give you options to debug or end. Press 'Debug' and the program will highlight for you the line of code that caused the error. Please note this line down and contact the first author, who will help you resolve the issue.

● TIMING

The following time estimates for each step of the procedure are based on two worksheets, each containing 5,000 data points, run in Excel 2010 on an Intel Core 2.20-GHz processor with 16 GB of RAM.

Steps 1–9: <1 min each

Step 10: 2–5 min

Steps 11–15: <1 min each

Step 16: 0.2–3 min, depending on the program

Steps 17 and 18: <1 min each

Steps 19–22: 10–20 min

ANTICIPATED RESULTS

The following subsections describe typical results obtained after running each code and further analysis that can be performed, if desired.

Plot_At_Origin

This program 'translates' each trajectory to the origin and plots it in a square chart along with all other trajectories of the same worksheet (**Fig. 5a**). The translated trajectories are displayed in the original input sheet in columns 8 and 9, under headings 'relative X' and 'relative Y'. Before plotting all the trajectories on the same sheet, the program goes through all sheets (i.e., conditions) and determines a reasonable size for the plot area, such that it will fit the longest trajectory. This may have a zooming-out effect. To zoom in manually, simply change the minimum and maximum of the axes. To zoom in programmatically, find line 'L = Round(1.1 * L/10) * 10' in the code of PlotAtOrigin_Main and change 1.1 to a smaller value (e.g., 0.6). Changing to a higher value will have the opposite effect of zooming out.

Make_Charts

This program plots each trajectory individually in its own separate chart, which is located next to the beginning of the xy coordinates data. The program asks the user to 'estimate the length of the edge of your plot area'. Enter a positive number, which corresponds to your guess of the length of the net displacement of the longest trajectory. After viewing the results, it will become apparent if you need to zoom in or to zoom out. This is the only program in DiPer that you can re-run over existing data. In order to zoom in, re-run Make_Charts with a small input number, and vice versa.

Speed

This program computes the average speed for each biological cell as average of its instantaneous speeds. Then it computes the grand average among all cells per condition. Finally, it plots the grand average as a column chart in sheet 'Speed by Cell', where error bars are the s.e.m. (**Fig. 5c**). If you wish to build a histogram of instantaneous speed, Excel provides this option in the Analysis Toolpack. Column 9 of the original sheet contains the instantaneous speed values that can be used as input for the Toolpack's histogram program.

Dir_Ratio

For each cellular trajectory, this program computes and displays the directionality ratio over elapsed time in the original input sheet in column 12. Next, all ratio distributions are aligned, and averages and s.e.m. are computed. These averages and s.e.m. are plotted as xy scatter lines in a new worksheet, called 'OverTime' (**Fig. 2a**). If the number of trajectories is low, the directionality curves will decay nonuniformly, exhibiting unexpected bulges that are due, at least in part, to the influence in speed. To obtain smoother curves, increase the cell number. This program also plots a bar chart for directionality ratio in the last point of the trajectory in a separate worksheet, called 'LastPoint'. This is provided to show the reader that this commonly used method is frequently not representative of average behavior, which is apparent in the OverTime curve.

MSD

This program performs the MSD analysis, as discussed in the INTRODUCTION. An input box will prompt you to enter the time interval between frames. The MSD coefficients are displayed on the original worksheets alongside the xy coordinates. A log-log plot appears in a new worksheet, called 'Graph', along with averages and s.e.m. listed in columns (**Fig. 2b**). If you wish to calculate the α -values, the relevant MSD coefficients for each biological cell are listed in rightmost region of the original worksheet. Use Excel's log and slope functions to calculate these. Please see the 'Limitations' section for a discussion of MSD and α -values. To lower computation time, the largest time interval is capped at $N/2$, that is, half the time length of the entire trajectory. To change this, find in the code the line 'Loop Until step $\geq (r_{last} - r_{top} - 1)/2$ ' and remove the $/2$ at the end. Doing so will extend the time interval to encompass all data. If you wish to lower the maximum time interval, you may replace the 2 with a higher number.

Autocorrel

An input box will prompt you to enter the time interval between frames and the desired number of time intervals for DA analysis. Please enter only numbers, and not units. The latter can be typed into the axis titles of the resulting chart. The resulting curves are displayed in a new worksheet, called 'Graph' (**Fig. 4a**). Each time point on the curve is the average autocorrelation coefficient for the given time interval, with error bars corresponding to s.e.m. These averages and s.e.m. are listed on the same worksheet, starting from lines 2 and 3, respectively. The autocorrelation coefficients used to compute averages are displayed on the original worksheets, where the user pasted his/her trajectory data, as well as in a worksheet called 'Stats', in which autocorrelation coefficients for different conditions are grouped per time interval. This format is easily amenable to statistical analysis in programs such as SigmaStat. These autocorrelation data are usually not normally distributed, so a nonparametric ANOVA based on ranks may be required to discern statistical difference between conditions.

If the curves decay too quickly or if they are very noisy, it may be a sign of oversampling, when centroid position changes rapidly from one frame to the other, rarely reporting productive displacements along the trajectory. Thus, you may wish to run the Sparse_Data program before running Autocorrel.

Autocorrel_NoGaps

This is a version of the program Autocorrel_Main. As discussed in the INTRODUCTION, this program precludes distance 'gaps' due to stops, that is, when xy coordinates do not change from one frame to the next. We recommend avoiding using this program if Autocorrel can be used in its place, as Autocorrel_NoGaps produces randomly oriented vectors in place of gaps.

This program will also ask you to input a distance threshold. This is the distance below which you consider the cell to be idling (i.e., not moving). Enter either zero or a small positive number, which should be a rather small percentage of the cell's physical size.

Autocorrel_3D

This program accepts 3D coordinates (xyz directions). If stops between frames occur, it will omit this nondisplacement and not make a DA coefficient. However, this is unlikely to occur, because 3D tracking is almost always automated.

Vel_Cor

The output for Vel_Cor is the same as that for Autocorrel: a new worksheet with the autocorrelation curves ('Graph') and a new worksheet with coefficients for each time interval ('Stats'). However, Vel_Cor does not ask you to input the number of time intervals over which to calculate the coefficients. Rather, Vel_Cor sets the number of intervals to equal one-third of the highest frame number. This is usually more than sufficient and prevents plotting the noise that often occurs in higher time intervals. The relevant line of code is: 'Do Until N >= (Ubound(arrX) - 2)/3.'

To change this programmatically to calculate all time intervals, simply delete the 3. To lower the threshold, you may increase 3 to a higher number, or you may simply go to the output chart and format the x axis to display the range you need.

Changing chart formatting programmatically

If you wish to change the formatting of charts programmatically, the text of the code can be changed. This is a good way to learn Visual Basic for Applications (VBA) and it can in the end save time. Go to the Visual Basic Editor (Alt+F11). In the text of the relevant program, find 'Sub Graphit ()'. This is a program that graphs your chart. **Table 2** shows a part of the code from Graphit in column 1 and the meaning in column 2.

In the program, you can change settings at will. For example, if you want the major unit to be 0.1 instead of 0.2, simply replace 0.2 with 0.1. For example, if you want the significant digits on the tick labels to go to two significant digits, replace 0.0 with 0.00 in the line 'TickLabels.NumberFormat = '0.0''. Once you make the required changes, rerun the new code on the Copy File, and NOT on your results.

TABLE 2 | Code from Graphit and meaning.

Code	Meaning
With .Axes(xlValue, xlPrimary)	With y axis
.Crosses = xlMinimum	Crosses x axis at the minimum
.MaximumScale = 1	Highest value on y axis equals 1
.MajorUnit = 0.2	This is the major unit
.TickLabels.NumberFormat = "0.0"	This is the number of significant digits
.HasTitle = True	y axis has a title
.AxisTitle.Orientation = xlUpward	...that is vertical and stretching upward
.AxisTitle.Text = "Speed"	The title of the axis is "Speed"
.AxisTitle.Font.Name = "Helvetica"	The font is Helvetica
.AxisTitle.Font.Bold = False	The title axis is NOT bold
.AxisTitle.Font.Size = 14	The font size of the title is 14
.TickLabels.Font.Size = 14	The tick labels font is also size 14
.AxisTitle.Left = 10	The title is offset 10 units to the left
End With	

Note: Any Supplementary Information and Source Data files are available in the [online version of the paper](#).

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